

INF1004 Mathematics II

Tutorial 5 Solutions: Singular Value Decomposition

SIT

1. Orthonormal Matrix Check

Question: Show that A is orthonormal.

$$A = \begin{pmatrix} 0.6 & -0.8 \\ 0.8 & 0.6 \end{pmatrix}$$

Recall: O is orthonormal if $O^T O = I$.

$$A^T = \begin{pmatrix} 0.6 & 0.8 \\ -0.8 & 0.6 \end{pmatrix}$$

$$\begin{aligned} A^T A &= \begin{pmatrix} 0.6 & 0.8 \\ -0.8 & 0.6 \end{pmatrix} \begin{pmatrix} 0.6 & -0.8 \\ 0.8 & 0.6 \end{pmatrix} \\ &= \begin{pmatrix} (0.36 + 0.64) & (-0.48 + 0.48) \\ (-0.48 + 0.48) & (0.64 + 0.36) \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \end{aligned}$$

Conclusion: Matrix A is orthonormal.

1. Orthonormal Matrix Check (Cont.)

Question: Show that B is orthonormal.

$$B = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \\ -1 & 0 & 0 \end{pmatrix}$$

$$\begin{aligned} B^T B &= \begin{pmatrix} 0 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \\ -1 & 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0+0+1 & 0+0+0 & 0+0+0 \\ 0+0+0 & 1+0+0 & 0+0+0 \\ 0+0+0 & 0+0+0 & 0+1+0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I \end{aligned}$$

Conclusion: Matrix B is orthonormal.

2. SVD of Rectangular Matrix (Step 1)

Given the matrix A :

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

- **Matrix Size:** 2×3 ($m = 2, n = 3$).
- **Condition:** Since $m < n$, we choose to diagonalize the smaller 2×2 matrix AA^T instead of the 3×3 matrix $A^T A$.
- **Goal:** Find $A = U\Sigma V^T$ starting with U .

Step 1: Compute AA^T and Eigenvalues

First, calculate the product AA^T :

$$AA^T = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

Solve the characteristic equation $|AA^T - \lambda I| = 0$:

$$\begin{vmatrix} 2 - \lambda & 0 \\ 0 & 1 - \lambda \end{vmatrix} = 0 \implies (2 - \lambda)(1 - \lambda) = 0$$

The eigenvalues are:

$$\lambda_1 = 2, \quad \lambda_2 = 1$$

Step 2: Singular Values and Σ

The singular values σ_i are the square roots of the eigenvalues:

$$\sigma_1 = \sqrt{2}, \quad \sigma_2 = \sqrt{1} = 1$$

Construct the 2×3 matrix Σ (matching the dimensions of A):

$$\Sigma = \begin{pmatrix} \sqrt{2} & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

Step 3: Finding Left Singular Vectors (U)

Find the normalized eigenvectors of AA^T for each λ :

- For $\lambda_1 = 2$: $\begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = 0 \implies u_2 = 0 \implies \vec{u}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$
- For $\lambda_2 = 1$: $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = 0 \implies u_1 = 0 \implies \vec{u}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Thus, the orthogonal matrix U is:

$$U = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Step 4: Finding Right Singular Vectors (V)

Compute v_i using $v_i = \frac{1}{\sigma_i} A^T u_i$:

- $v_1 = \frac{1}{\sqrt{2}} A^T \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

- $v_2 = \frac{1}{1} A^T \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$

To find v_3 (orthogonal to v_1, v_2), solve $Av_3 = 0$:

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies y = 0, x = -z \implies \vec{v}_3 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

Final Result: V^T Matrix

The complete V matrix is formed by $[v_1, v_2, v_3]$:

$$V = \begin{pmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \end{pmatrix}$$

The SVD components for $A = U\Sigma V^T$ are now complete.

3. SVD of Symmetric Matrix

$$A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$$

1. Find $A^T A$:

$$A^T A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 10 & 6 \\ 6 & 10 \end{pmatrix}$$

2. Find eigenvalues λ :

$$(10 - \lambda)^2 - 36 = 0 \implies (10 - \lambda) = \pm 6 \implies \lambda = 16, 4$$

Singular values: $\sigma_1 = 4, \sigma_2 = 2$.

$$\Sigma = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}$$

3. SVD of Symmetric Matrix (Cont.)

3. Find Eigenvectors (Normalized V): **For** $\lambda_1 = 16$:

$$\begin{pmatrix} -6 & 6 \\ 6 & -6 \end{pmatrix} \vec{x} = 0 \implies \vec{v}_1 = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

For $\lambda_2 = 4$:

$$\begin{pmatrix} 6 & 6 \\ 6 & 6 \end{pmatrix} \vec{x} = 0 \implies \vec{v}_2 = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$$

4. Find U using $u_i = \frac{1}{\sigma_i} A v_i$:

$$u_1 = \frac{1}{4} \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 4/\sqrt{2} \\ 4/\sqrt{2} \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

$$u_2 = \frac{1}{2} \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2/\sqrt{2} \\ -2/\sqrt{2} \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$$

Solution 3: Final SVD Answer

For the symmetric matrix $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$:

$$A = U\Sigma V^T$$

$$\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} = \underbrace{\begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}}_U \underbrace{\begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}}_\Sigma \underbrace{\begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}}_{V^T}$$

Note: Since A is symmetric and positive definite, $U = V$ and the singular values are exactly the eigenvalues.

4. SVD of Singular Matrix

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

1. Find $A^T A$:

$$A^T A = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$$

Characteristic equation: $(2 - \lambda)(2 - \lambda) - 4 = 0$, solve to get Eigenvalues:
 $\lambda_1 = 4, \lambda_2 = 0$. Singular values: $\sigma_1 = 2, \sigma_2 = 0$.

2. Find V (Normalized Eigenvectors): $\lambda_1 = 4 \implies \vec{v}_1 = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$.

$$\lambda_2 = 0 \implies \vec{v}_2 = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}.$$

3. Find U :

$$u_1 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2/\sqrt{2} \\ 2/\sqrt{2} \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

$$u_2 \text{ must be orthonormal to } u_1, \text{ so } u_2 = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}.$$

Solution 4: Final SVD Answer

For the singular matrix $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$:

$$A = U\Sigma V^T$$

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \underbrace{\begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}}_U \underbrace{\begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}}_{\Sigma} \underbrace{\begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}}_{V^T}$$

Note: The zero on the diagonal of Σ reflects that the matrix is rank 1.

5. SVD of 3x4 Matrix (Step 1)

$$A = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

Note: For wide matrices, it is faster to use AA^T to find U first.

1. Find AA^T (3x3):

$$AA^T = \begin{pmatrix} 2 & 0 & 2 \\ 0 & 2 & 0 \\ 2 & 0 & 2 \end{pmatrix}$$

2. Find eigenvalues of AA^T : $\lambda_1 = 4, \lambda_2 = 2, \lambda_3 = 0$.

$\sigma_1 = 2, \sigma_2 = \sqrt{2}, \sigma_3 = 0$.

$$\Sigma = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & \sqrt{2} & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

5. SVD of 3x4 Matrix (Step 2)

3. Find U (Normalized Eigenvectors of AA^T):

For $\lambda = 4$: $\vec{u}_1 = \begin{pmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{pmatrix}$. For $\lambda = 2$: $\vec{u}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$. For $\lambda = 0$:

$$\vec{u}_3 = \begin{pmatrix} 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \end{pmatrix}.$$

$$U = \begin{pmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \end{pmatrix}$$

5. SVD of 3x4 Matrix (Step 3)

4. Find V using $v_i = \frac{1}{\sigma_i} A^T u_i$:

$$v_1 = \frac{1}{2} A^T u_1 = \frac{1}{2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \\ 0 \end{pmatrix}$$

$$v_2 = \frac{1}{\sqrt{2}} A^T u_2 = \begin{pmatrix} 0 \\ 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{pmatrix}$$

For null space (since $\lambda_3 = 0$), we find vectors orthogonal to v_1, v_2 :

$$v_3 = \begin{pmatrix} 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \\ 0 \end{pmatrix}, \quad v_4 = \begin{pmatrix} 0 \\ 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \end{pmatrix}$$

Final Answer: $A = U \Sigma V^T$.